nous on MemoryAgentBench: A Paper-Faithful Evaluation of a Cognitive Memory Agent

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Abstract

We evaluate nous, a cognitive memory agent with episodic ingestion, LLM-based fact extraction, and sleep-cycle consolidation, on MemoryAgentBench (Hu et al., ICLR 2026), a benchmark that measures four memory competencies of conversational agents: Accurate Retrieval (AR), Test-Time Learning (TTL), Long-Range Understanding (LRU), and Conflict Resolution (CR). We reproduce the benchmark's evaluation protocol exactly—its per-dataset answer prompts, its per-dataset graders (including two GPT-4o judge protocols), vendored from the benchmark's open-source repository and verified line-by-line by an independent review—and drive a real nous HTTP server through its full production cognitive cycle for every test instance. On the evaluated subsets, nous scores **AR 0.897** (n=232, 95% CI [0.86, 0.94]), **CR 0.766** (n=64), **LRU 0.824** (n=68, CI [0.73, 0.91]), and **TTL 0.555** (n=200, CI [0.49, 0.62]). The first three are robustly above every method in the benchmark paper's results (best per-competency averages there: 71.8%, 29.5%, 62.2%); TTL sits at the top of that range (53.9%). Against the 2026 field (full-benchmark results reported by Infini Memory with a gpt-5-mini base), the honest placement is: above the best reported Accurate Retrieval, second on Conflict Resolution, comparable on Long-Range Understanding, and behind on Test-Time Learning. We describe the testing technique, a statistical-validation pass that materially corrected one competency's initial estimate, and the threats to validity that qualify a direct leaderboard comparison.

1 Introduction

Long-running LLM agents need memory that outlives a single context window: they must store what they were told, learn new skills from it, reason over inputs far larger than the model context, and revise beliefs when information changes. MemoryAgentBench [1] operationalizes these requirements as four competencies and evaluates them by streaming large corpora into an agent incrementally and then asking questions against what the agent retained.

nous is a cognitive AI agent whose memory subsystem combines an episodic store (conversation episodes and summaries), a semantic store (LLM-extracted facts with typed graph edges), full-fidelity content chunks with hybrid (vector + lexical) retrieval, and an offline *sleep* consolidation cycle that links, deduplicates, and densifies memory between sessions. This report documents a from-scratch evaluation harness that measures nous on MemoryAgentBench under its *production* configuration, and places the results against the methods evaluated by the benchmark's authors.

Two principles governed the effort:

1. **Paper fidelity.** The score must be computed exactly as the benchmark computes it—same answer prompts, same graders, same judge models—so numbers are comparable. All grading

code was vendored from the benchmark's MIT-licensed repository and cross-reviewed against it by an independent reviewer agent with no stake in the outcome.

2. **Statistical honesty.** No above/below-field claim is made unless the 95% confidence interval of the measured accuracy clears the best published field value. This rule changed one headline conclusion (Section 4.3).

2 System Under Test

2.1 The nous cognitive cycle

Each benchmark instance exercises the complete production write path:

1. **Ingestion.** The instance's context is split into chunks of at most 32,000 characters (conversational corpora are packed turn-by-turn into chunks of the same size, preserving every turn) and streamed to the agent as consecutive chat turns in one session.

2. **Session close.** The session is closed, triggering the durable-write chain: an episode summarizer, an LLM fact extractor that emits typed facts, and full-fidelity chunk embedding.

3. **Sleep consolidation.** One or more sleep cycles run offline consolidation: fact deduplication, cross-episode linking, co-occurrence edge formation, and graph densification.

4. **Answering.** Every question is asked in a *fresh* session. The agent answers from its memory (retrieval-augmented context injection plus agentic recall tools); the question is framed with the benchmark's own per-dataset prompt template.

2.2 Configuration

nous ran under its live production configuration (Claude Sonnet 5 as the reasoning model; hybrid chunk search; typed-edge extraction; cross-encoder re-ranking; date-aware retrieval; adaptive thinking) with exactly three eval-hygiene overrides, all disabling background self-improvement features irrelevant to memory measurement: the heartbeat timer, the scheduler, and rubric outcome detection. A 117-knob snapshot of the configuration is pinned in the harness repository for reproducibility.

Two run profiles were used. The *full-fidelity* profile (unbounded episode summarization, three sleep cycles) is used for moderate contexts. For multi-megabyte contexts a *big-context* profile bounds the episode summarizer to 8 chunks, runs one sleep cycle, and paces ingestion turns (15 s); this profile eliminated upstream rate-limit stalls and ingest timeouts entirely (0 errors across all final runs, including 2.3M-character documents).

2.3 Isolation

Every instance gets a fresh nous server process, a unique agent identity, and a dedicated evaluation database; nothing carries over between instances. Control probes confirmed the isolation: on Conflict Resolution, the identical questions asked of an agent *before* ingestion score 0.000, so the measured accuracy is attributable to memory, not to model priors.

3 Benchmark and Datasets

MemoryAgentBench streams each dataset's context into the agent and then asks held-out questions. Table 1 lists the evaluated subsets.

Competency	Source	Context size	Questions used	Grader
CR	factconsolidation_sh (6k–262k)	26k–1.1M chars	4×8	substring (SQuAD-norm.)
	factconsolidation_mh (6k–262k)	26k–1.1M chars	4×8	substring (SQuAD-norm.)
	single-hop (sh) = updated fact; multi-hop (mh) = compose two updated facts			
AR	eventqa_65536	~285k chars	200 (5 ctx × 40)	all-elements recall
	eventqa_131072	~570k chars	8	all-elements recall
	eventqa_full	~2.3M chars	8	all-elements recall
	ruler_qa1_197K / qa2_421K	0.9M / 1.9M chars	8+8	substring
	longmemeval_s*	1.6M chars (1282 turns)	8	GPT-4o yes/no judge
TTL	icl_banking77 / clinic150 / nlu	~400k chars each	3×40	parsed-label exact match
	icl_trec_coarse / trec_fine	~400k chars each	2×40	parsed-label exact match
LRU	detective_qa (10 novels)	~400k chars each	68	substring
	infbench_sum_eng	1.75M chars	1	GPT-4o 3-call f1 judge

Table 1: Evaluated datasets. "Context size" is the raw document streamed into memory. The TTL *recsys* sub-dataset could not be graded faithfully: its grader requires an entity-id mapping file that is not shipped in the benchmark's public repository, so TTL here is the in-context-learning family only.

4 Testing Technique

4.1 Paper-faithful grading

Each sub-dataset is graded exactly as in the benchmark's `post_process` dispatch:

- **Substring match** (ruler, detective, factconsolidation): the SQuAD-style normalization (lowercase, strip punctuation and articles, collapse whitespace), then gold-in-prediction containment, max over golds.
- **EventQA recall**: *raw* case-insensitive containment of every gold element (`el.lower() in pred.lower()`); the answer is correct only if all elements are present. An initial implementation that reused the SQuAD normalization here was flagged by independent review as strictly more lenient than the paper (punctuation and article stripping create false positives) and was corrected before any reported run.
- **ICL exact match**: parse the label the prompt requests (`label: X`), then exact match on the bare label. This is a deliberate, conservative interpretation: the benchmark's own ICL scoring parses with a different prefix and reports multiple metrics of which the substring variant false-positives on digit collisions (gold `5` matches `label:  15`); our parser cannot.
- **LongMemEval judge**: the benchmark's GPT-4o yes/no judge with its verbatim per-question-type prompts (including the temporal-reasoning off-by-one tolerance and the abstention protocol for `_abs` questions).

- **Summarization judge** (InfBench-Sum): the benchmark's three-call GPT-4o protocol scoring fluency $f \in \{0, 1\}$, key-point recall r, and sentence-level precision p, combined as $\text{f1} = f \cdot \frac{2rp}{r+p}$.

Answer prompts are likewise the benchmark's own per-dataset templates (e.g., the CR template that states the serial-number update rule with few-shot examples; the eventQA template that frames "what happened next"). On identical memory, switching from a generic answer prompt to the benchmark's CR template moved CR accuracy by +28 percentage points—prompt fidelity is not optional if numbers are to be compared.

4.2 Independent review

All vendored grading and prompting code was audited by a reviewer agent with fresh context against the benchmark's repository, file by file. The review confirmed byte-fidelity of the normalizers, judge prompts, dispatch tables, and the summarization formula; it found exactly one fidelity bug (the eventQA normalization above), which was fixed and regression-tested before the reported runs. The harness carries 125 unit tests covering graders, judges, chunking, persistence, and isolation.

4.3 Statistical validation

Initial runs used 8 questions per source. A dedicated validation pass then grew the sample until each competency's 95% CI either cleared the best published field value or demonstrably did not:

- **TTL corrected downward.** At n=8/source TTL measured 0.650; at n=40/source (n=200) it fell to **0.555** (CI [0.486, 0.624]), with one source falling from 0.625 to 0.450. The small sample had been an optimistic draw. TTL is therefore reported as *at* the field ceiling, not above it.
- **LRU confirmed.** Detective-QA at all 10 novel contexts (n=68) held at **0.824** versus 0.833 at n=6; its lower CI bound (0.733) clears the best field value (0.622).
- **AR firmed upward.** EventQA-65536 across 5 contexts at n=200 rose to 0.910 versus 0.875 at n=24, consolidating AR at **0.897** (CI [0.857, 0.936]).

The general lesson—small-sample memory-eval estimates run optimistic; grow n until the CI clears the comparison band—is recorded as a standing methodological rule for this project. All results persist per-question to append-only JSONL as each instance completes, so partial runs are never lost and every reported number is recomputed from the persisted records.

5 Results

5.1 nous scores

Per-source detail: CR single-hop 0.906 vs. multi-hop 0.625 (the benchmark reports at most ∼7% for multi-hop across all its methods); AR eventqa 0.910/0.750/0.875 (65k/131k/full), ruler 0.875/0.750 (197K/421K), longmemeval 0.875; TTL banking77 0.525, clinic150 0.500, nlu 0.425, trec-coarse 0.875, trec-fine 0.450; LRU detective 56/68.

Competency	Score	n	95% CI	Best published avg. / verdict
Accurate Retrieval	**0.897**	232	[0.857, 0.936]	0.718 — **above, robust**
Conflict Resolution	**0.766**	64	[0.66, 0.87]	0.295 — **above, robust**
Long-Range Understanding	**0.824**	68	[0.733, 0.914]	0.622 — **above, robust**
Test-Time Learning	**0.555**	200	[0.486, 0.624]	0.539 — at ceiling

Table 2: nous on MemoryAgentBench (paper-faithful grading, production configuration, 0 execution errors on all final runs). LRU is detective-QA; the auxiliary InfBench-Sum summarization judge scored f1 = 0.592 on a single 1.75M-char book (unfirmed, n=1). "Best published avg." is the strongest per-competency average across all methods in the benchmark's results table. *Comparability tiers:* AR, CR, and LRU are **T1** (paper's exact prompt/grader/questions; sole deviation is ingestion transport, Section 6); TTL is **T2** (T1 plus a deliberate, conservative label-parsing choice in the ICL grader). No single aggregate score is published: TTL omits recsys (grader data file not publicly shipped), and the giant AR/LRU sources are measured at 1 context each. The repository's `BASELINE_SUMMARY.md` reconciles every number here to its committed per-question evidence file, tier, and missing-components list.

5.2 Comparison with the benchmark paper's field (2025)

Table 3 reproduces the per-competency averages of representative methods from the benchmark's results table (its Table 3; the fourth competency is named Selective Forgetting there and Conflict Resolution in earlier versions—same datasets). These rows date from the benchmark paper; the 2026 field is covered in Section 5.3.

Method	Class	AR	TTL	LRU	CR/SF
GPT-4o-mini (full context)	long-context	49.2	48.6	46.2	25.0
GPT-4.1-mini (full context)	long-context	71.8	46.2	49.1	20.5
Claude-3.7-Sonnet (full context)	long-context	59.7	**53.9**	**62.2**	22.5
BM25	RAG	60.5	44.5	35.6	25.5
Text-Embed-3-Large	RAG	54.6	44.3	37.3	16.0
HippoRAG-v2	RAG	65.1	35.8	36.2	**29.5**
Mem0	agentic memory	32.6	21.2	20.7	10.0
GraphRAG	structure-aug.	40.9	24.8	19.9	8.0
MIRIX	agentic memory	47.5	24.1	25.4	8.0
nous (this work)	cognitive memory	**89.7**	55.5	**82.4**	**76.6**

Table 3: Per-competency averages (%). Field rows are from the benchmark's published results; bold marks the best value per column among field methods, and nous where it exceeds all of them. Comparison caveats in Section 6 apply—in particular the base-model mismatch and the subset sizes.

Three observations against the paper's table:

1. **Conflict Resolution is the standout vs. the 2025 field.** nous scores 0.766 where no method in the paper exceeds 0.295, and 0.625 on multi-hop conflict composition where that field is at most ∼0.07. We attribute this to nous's explicit fact store with update semantics: conflicting facts are extracted, linked, and consolidated during sleep, rather than left as contradictory passages for retrieval to stumble over.

2. **Retrieval and long-range understanding benefit from full-fidelity chunk memory.** AR 0.897 and LRU 0.824 exceed both the RAG family and the long-context family, including on 1.6–2.3M-char inputs that no context window holds.

3. **Test-time learning is the honest gap.** TTL 0.555 edges the best value in the paper (0.539) but the CI straddles it. Learning a classification skill from thousands of in-context examples is not what an episodic/semantic memory is optimized for; accuracy here tracks how many labeled exemplars retrieval can surface per query.

5.3 Comparison with the 2026 field

The field has moved since the benchmark paper. Infini Memory [2] (2026) re-evaluated a modern method suite on the full benchmark with gpt-5-mini as the base model, 4096-token ingestion chunks, and a gpt-5 LLM-as-judge protocol. Table 4 reproduces its results.

Method (gpt-5-mini base)	AR	TTL	LRU	CR/SF
Mem0	28.0	36.0	42.0	35.5
RAPTOR	32.0	39.0	38.2	48.2
MemoRAG	35.0	36.0	25.0	54.4
MemGPT	49.0	30.0	42.7	45.6
LightMem	57.0	34.0	69.2	39.2
REMem	62.0	42.0	62.3	41.2
HippoRAG-v2	78.0	70.0	54.7	72.2
Infini Memory-H	77.0	76.0	76.0	83.0
Infini Memory-A	**83.0**	**79.0**	**79.3**	**83.6**
nous (this work, slices)	**89.7**	55.5	82.4†	76.6

Table 4: The 2026 field, as reported by Infini Memory [2] (full benchmark; per-competency averages, %). Cross-comparison caveats beyond Section 6: the 2026 numbers use a gpt-5 *LLM-judge for all tasks* rather than the benchmark's official per-dataset graders (LLM judges are typically more lenient than substring matching), their TTL includes recsys and their LRU includes full summarization (ours do not), and our numbers are slices, not the full benchmark. †detective-QA only.

The honest read against the 2026 frontier: nous's AR slice (0.897) remains above the best reported value (0.830); CR (0.766) is second to Infini Memory-A (0.836) and above everything else; LRU (0.824, detective only) is comparable to the leader (0.793, incl. summarization); and **TTL (0.555) is clearly behind the 2026 leaders** (0.70–0.79, though those include recsys and a more lenient judge). The 2025-table margins (Table 3) substantially reflect the base-model generation; against contemporaries, nous is competitive-to-leading on retrieval and conflict handling and behind on test-time learning. A footnote on aggregates: Infini Memory's own abstract reports 64.7% overall while its results table's competency average is 81.2% — aggregation-method ambiguity even within one paper, which is why we publish no aggregate.

6 Threats to Validity

1. **Base-model mismatch.** nous reasons with Claude Sonnet 5; the benchmark's agentic-memory rows mostly use GPT-4o-mini, and its strongest long-context rows use GPT-4.1-mini

and Claude-3.7-Sonnet. Part of nous's margin is attributable to the stronger base model. The long-context Claude-3.7 row (best field TTL and LRU) is the fairest anchor, and nous still exceeds it on AR (+30pp), LRU (+20pp), and CR (+54pp).

2. **Subsets, not the full benchmark.** Question counts per source range from 8 to 200 (Table 1); the giants (eventqa-full, ruler-421K, longmemeval) were measured at one context each. CIs in Table 2 reflect this. recsys (TTL) is excluded for lack of the grader's data file; InfBench-Sum is a single book.

3. **Configuration provenance.** The production configuration was improved using failures observed on this benchmark's CR subset in an earlier phase (write-path fidelity, hybrid chunk retrieval, recall depth). The evaluation is therefore best read as "nous after its memory bugs were fixed," not as a blind first attempt. No knob was tuned per-dataset, and the same configuration serves all four competencies.

4. **Ingestion transport differs from the paper's protocol.** The benchmark feeds context in chunks of 512 tokens for the doc-QA, LongMemEval, and conflict-resolution tasks and 4096 tokens elsewhere (uniformly 4096 for the commercial memory agents, for cost); our harness streams 32,000-character turns, and nous re-chunks internally for retrieval, so transport granularity is not the retrieval granularity. The baseline is therefore paper-faithful in prompts, graders, judges, and question sets, but *not* an exact reproduction of the paper's ingestion transport. No answer-bearing text was lost to this difference: ingest capacity was 80 $\times$ 32k chars (2.56M), and a post-hoc audit of all 34 evaluated instances at the exact run settings confirmed **zero dropped chunks** (largest instance: 2.31M chars, 73/80 chunks; the 1,282-turn LongMemEval conversation packed to 50 chunks with every turn retained). Runs now stamp `chunks_sent/chunks_truncated` on every persisted result row, and truncated instances are excluded from headline aggregates by construction.

5. **Benchmark prompts embed task rules.** The benchmark's own CR and eventQA templates tell the agent the update rule / task framing. We use them because the field numbers use them; they help any system.

6. **Judge sharing.** LongMemEval and InfBench-Sum scores depend on GPT-4o judges, as in the benchmark.

7. **The 2026 comparison is as-reported, not re-run.** Table 4 takes the 2026 numbers from the Infini Memory paper [2] at face value; we did not reproduce them. Their protocol differs from the benchmark's official grading (gpt-5 LLM-judge for all tasks, typically more lenient than substring graders; 4096-token chunks) and from ours, and their coverage is the full benchmark where ours is slices. Per-competency comparison is indicative, not exact; we make no state-of-the-art claim.

7 Conclusion

On the sampled, paper-aligned slices and under the benchmark's exact grading, nous exceeds every method reported in the MemoryAgentBench paper on three of four memory competencies—Accurate Retrieval (0.897), Conflict Resolution (0.766, a 2.6$\times$ margin over that table's best), and Long-Range Understanding (0.824)—and matches that table's best Test-Time Learning (0.555). Against the 2026 field (Section 5.3, as reported by Infini Memory with a gpt-5-mini base and a more lenient judge protocol), the honest placement is: **leading on retrieval** (AR slice above the best 2026 value), **second on conflict resolution** (0.766 vs. 0.836), **comparable on long-range understanding**, and **behind on test-time learning**. None of this is a state-of-the-art or official-benchmark claim (Section 6, items 2 and 6). The conflict-resolution result still suggests that an explicit extract-link-consolidate memory architecture addresses a failure mode most retrieval systems share, and the clear TTL gap identifies where `nous` should improve next; both justify a full-benchmark evaluation. The harness, vendored graders, pinned configuration, and per-question result records are available in the project repository; `BASELINE_SUMMARY.md` reconciles every number to its evidence file and comparability tier.

References

[1] Y. Hu et al., "Evaluating Memory in LLM Agents via Incremental Multi-Turn Interactions" (MemoryAgentBench), ICLR 2026. `github.com/HUST-AI-HYZ/MemoryAgentBench`, arXiv:2507.05257.

[2] "Infini Memory: Maintainable Topic Documents for Long-Term LLM Agent Memory," arXiv:2606.10677, 2026.